



SniperSheet

AI-Powered Excel Formula Engine

Natural Language → Precise Excel Formulas in Seconds

35+
Formula Patterns

4
Smart Tabs

100%
Free AI

3
AI Models

Smart Hub

Commands

Cell Dimensions

Advanced Tools

Local Fallback

100% Free





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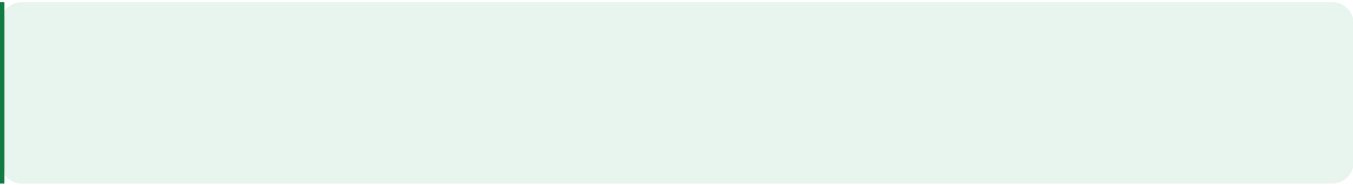
Complete 17-page official guide

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What is SniperSheet?

AI-powered Excel add-in for natural language formula generation



Groq AI Engine
Llama 3.3 70B
100% free, no account needed

Bilingual Support
English & Arabic input
Full RTL interface

Proxy Security
US Replit server proxy
No VPN required globally

Offline Fallback
35+ built-in patterns
Zero-latency local engine

Office.js Add-in
ReadWriteDocument access
Excel Ribbon integration

400px Task Pane
Smooth scrolling UX
Native Excel sidebar



Who is SniperSheet For?

Target professionals and primary use cases

Engineers
Complex formulas in seconds

Analysts
Financial & KPI modeling

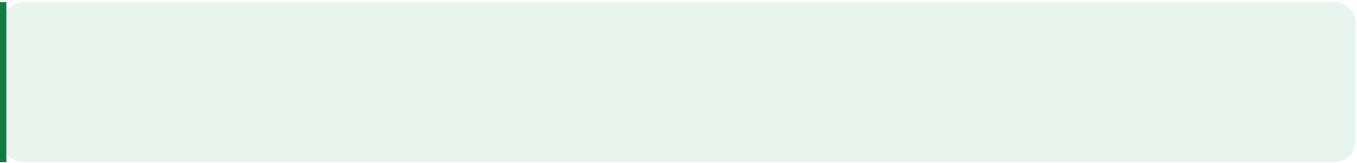
Project Managers
Tracking & budgeting

Students
Stats & research analysis



Task Pane Interface

400px wide · 4-tab navigation · bilingual RTL/LTR support



01

Smart Hub

- AI formula generation
- Confidence score 0–
- Formula type detection
- Word Radar type check
- Style & formatting hints
- History log with

02

Commands

- Arabic command aliases
- SUM. AVG. MAX. MIN.
- BONUS 15%. TAX 15%
- Real-time calculation
- Searchable reference
- Full execution history

03

Cell Dimensions

- Arabic character width
- Font size & bold
- Single & batch modes
- Pixel and Excel unit
- Recommended padding
- One-click copy results

04

Advanced Tools

- Empty Field Radar scan
- Smart Print-Fit calc
- Professional Report
- CSV / tab-data input
- Color-coded grid output
- Browser print dialog



Design System

Color palette and typography

Primary Green #107C41	Dark Green #0B5C30	Light Green #E8F5EE	Neutral 900 #111827	Neutral 600 #4B5563	Neutral 200 #DDE3EA



Smart Hub — AI Formula Engine

Tab 1 · Natural language ↔ Excel formula via Groq AI



Full Feature List

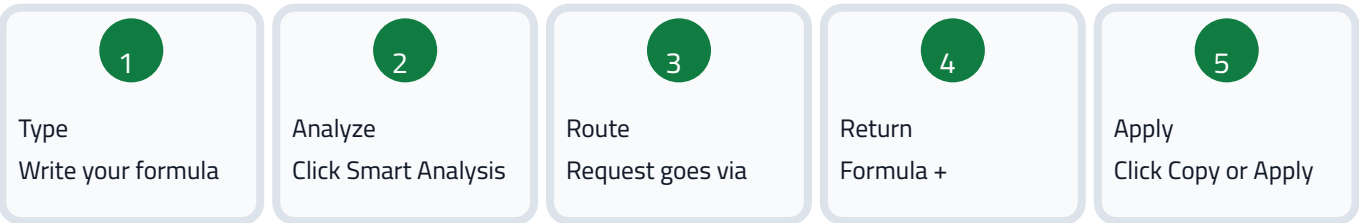
Everything Smart Hub offers

- AI formula generation (Groq Llama 3.3 70B)
- Confidence scoring: 0–100% per formula result
- Formula type detection (IF, XLOOKUP, SUM...)
- Word Radar: flags typos & ambiguous terms
- Style hints: cell color, bold, italic formatting
- History log with timestamps and replay
- Local formula engine — full offline fallback
- 3-model AI cascade with auto-retry on failure
- Status badge: success / warning / error state
- Step-by-step formula explanation provided
- One-click copy formula to clipboard
- One-click apply formula to active Excel cell



How It Works

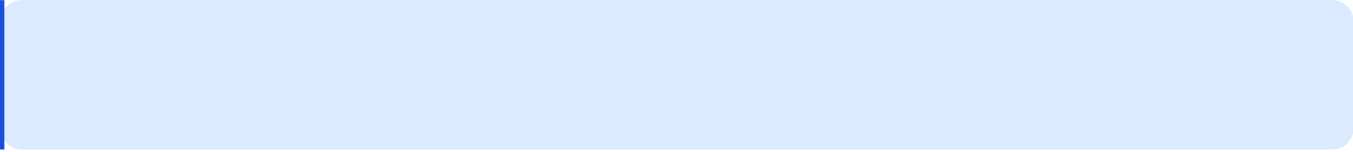
Step-by-step: description to formula in seconds





Commands — Formula Aliases

Tab 2 · Arabic & English shorthand for instant calculations



Command (EN / AR)	Description	Formula
SUM / جمع	Sum of all values in range	=SUM(B1:B10)
MULTIPLY / ضرب	Product (multiply) of values	=PRODUCT(A1:A5)
AVERAGE / متوسط	Arithmetic mean of a range	=AVERAGE(D1:D20)
MAX / أكبر	Maximum value in range	=MAX(C1:C50)
MIN / أصغر	Minimum value in range	=MIN(C1:C50)
COUNT / عدد	Count of numeric cells	=COUNT(E1:E100)
BONUS / مكافأة	Adds 15% bonus to amount	=A1*1.15
TAX / ضريبة	Calculates 15% tax	=A1*0.15
PERCENTAGE / نسبة	Percentage of value vs total	=(A1/B1)*100
IF / إذا	Simple conditional check	=IF(A1>1000,"High","Low")
COUNTIF / شرط عدد	Count cells matching condition	=COUNTIF(B:B,">=60")
SUMIF / شرط جمع	Sum cells matching condition	=SUMIF(C:C,">1000";D:D)

Supported Features

- Searchable command reference table
- Real-time result calculation
- Execution history with timestamps
- Combine commands: BONUS + SUM

Usage Examples

```
"SUM B1:B10" !' =SUM(B1:B10)
"BONUS A1" !' =A1*1.15
"bÆEc' # ٠## " " 05T0,,# ٠## •
"PERCENTAGE A1 B1" !' =(A1/B1)*100
```



Cell Dimensions Calculator

Tab 3 · Optimal cell sizing for Arabic & English text



How It Works

Input parameters → algorithm → Excel unit output

Text Content

Enter cell text. Supports Arabic, English, or mixed.

Font Size (pt)

Specify font size in points. The calculator adjusts

Bold & Italic

Toggle bold (×1.15) or italic (×1.05) multipliers.

Padding

Specify padding in pixels per side. Added to final



Output Formats

What the calculator returns and available actions

- Width in pixels (px)
- Width in Excel column units
- Height in pixels (px)
- Height in Excel row units
- Recommended padding values
- Character count & text metrics
- Single cell calculation mode
- Batch: multiple cells at once
- One-click copy to clipboard
- Apply directly to active cell
- Save favorite settings
- Arabic / English / mixed aware





Advanced Tools — 3 Utilities
Tab 4 · Empty Radar, Smart Print-Fit, Professional Report

Empty Field Radar

Data Validation

- CSV or tab-separated data input
- Visual grid: green=filled, red=empty
- Exact coordinate list (Row:Col)
- Auto header row detection
- Filled vs empty count summary

Smart Print-Fit

Print Layout

- A4 and A3 paper support
- Portrait & landscape orientation
- Excel column width unit output
- Recommended font size output
- Print-ready in under 3 clicks

Professional Report

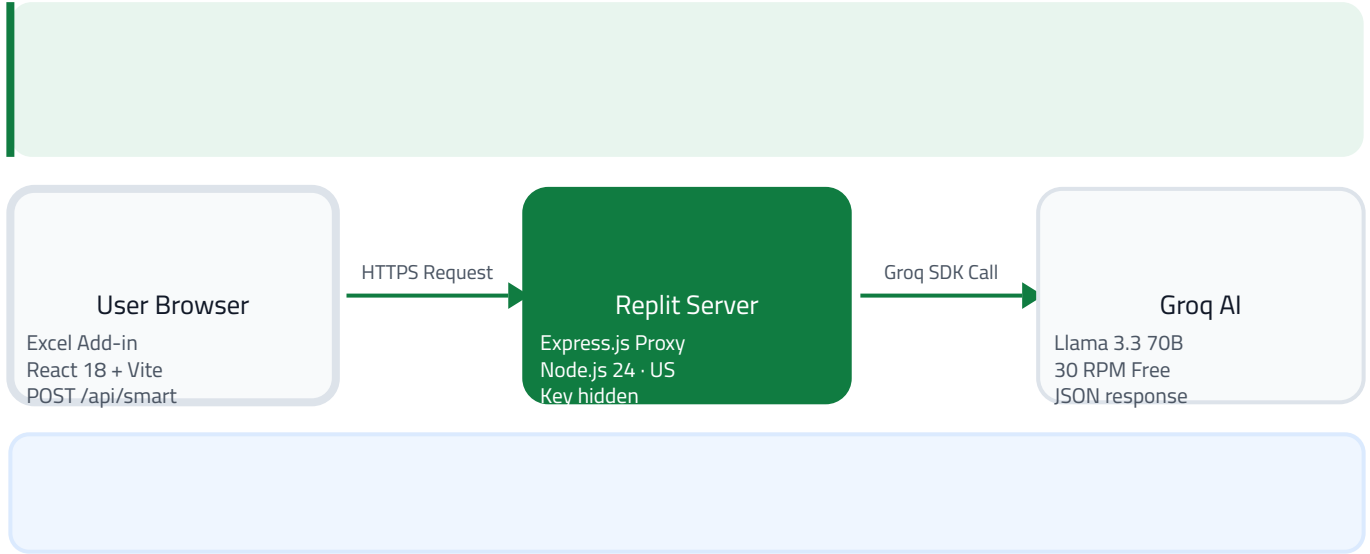
Reporting

- Styled HTML from raw data
- Alternating row color scheme
- Bold column headers auto-styled
- Auto header row detection
- One-click browser print trigger



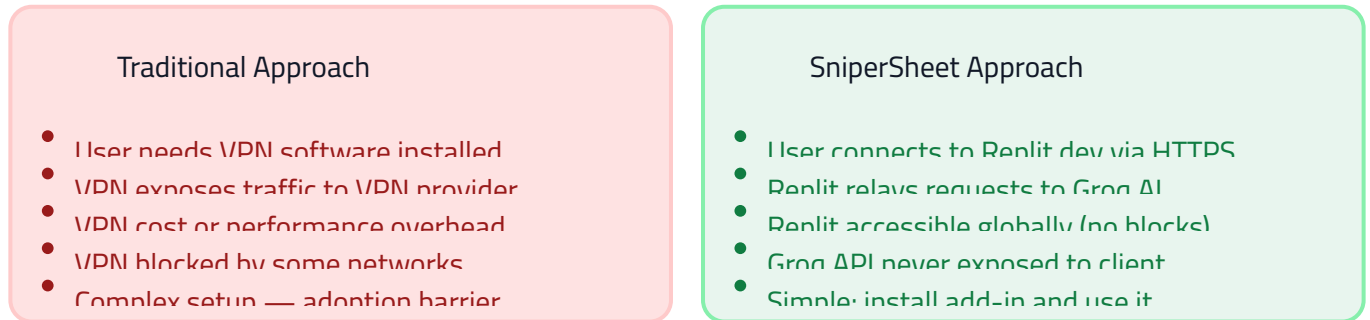
AI Proxy Architecture

How SniperSheet securely connects users to Groq AI via Replit



No-VPN Design

Why users in Iraq and restricted regions can use SniperSheet





Groq AI

The AI Assistant that's 100% Free

FREE FOREVER
No billing · No user account



3-Model Cascade

SniperSheet tries models in order — automatic retry

PRIMARY	FALLBACK	BACKUP	OFFLINE
llama-3.3-70b-versatile	llama3-70b-8192	llama3-8b-8192	Local Formula Engine
70B32K ctx 30 RPM	70B8K ctx 30 RPM	8B8K ctx 30 RPM	— ∞
Best quality · Latest Llama 3.3	Proven stable · Activates if primary	Ultra-fast · For simple formula	35+ patterns · Zero latency ·



Groq LPU Technology

Purpose-built hardware for ultra-low-latency AI inference

~1–3s

30 RPM

\$0

99%+



Replit

AUTOSCALE
Always-on - US region



Server Stack

Complete technical profile of the SniperSheet backend

Property	Value	Notes
	Node.js	Latest LTS — full ESM support
	Express	Asvnc/await middleware
	ts-node	TypeScript native npm package
	tsconfig-paths	Full type safety end-to-end
	fastify	Fast CIS bundle for production
	monorepo	Monorepo-compatible, fast installs
	serverless	Scales to zero when idle
	aws	Low latency — accessible globally
	https	All traffic encrypted
	whitelist	Add-in domain whitelisted



API Endpoints

Routes served by the SniperSheet Express server

POST	/api/smart/analyze	Main AI formula generation — accepts {prompt, lang}
GET	/api/addin/manifest.xml	Office.js manifest for Excel add-in sideloading
GET	/api/addin/icon-*.png	PNG icons: 16, 32, 64, 80px for the Ribbon
GET	/excel-addin/	React task pane SPA (Vite production build)



Formula Examples Gallery
12 formulas generated by Smart Hub AI from natural language

Natural Language Input	Formula Generated	Category

Legend:

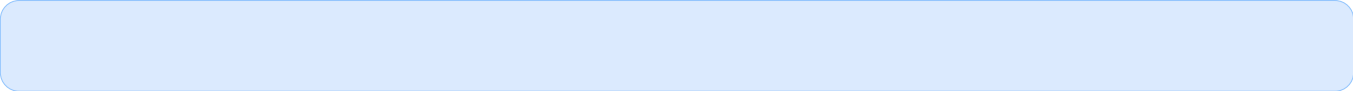
Conditional

Lookup

Statistical

Financial

Date





Writing Effective Prompts

How to describe your calculation for the best AI results



Prompts That Work Well

Examples of effective formula descriptions

	Score: 96%
Why: Names cells (A1, B1), states the condition (>40), and the	=IF(A1>40,(A1-40)*B1*1.5,0)
	Score: 94%
Why: Names both columns, defines the lookup key and the return	=XLOOKUP(F2,A:A,C:C,"Not found")
	Score: 97%
Why: Names both columns and clearly separates the condition column	=SUMIF(C:C,">1000",D:D)
	Score: 99%
Why: States all grade boundaries explicitly and in the correct order for	=IFS(B2>=90,"A+",B2>=75,"B",TRUE,"C")



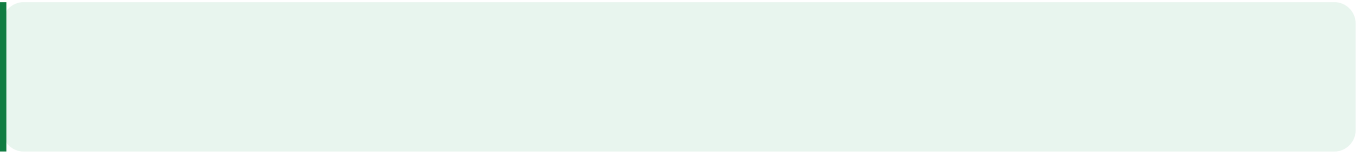
Prompts to Avoid

Vague descriptions that produce low confidence



Confidence Score System

How SniperSheet measures and displays formula accuracy



85–100%

High Confidence

Formula is highly likely to be correct. Prompt was clear and specific. Use as-is.

- Named specific columns or cells
- Condition values stated explicitly
- Formula type clearly implied
- No Word Radar warnings

Score 97%: "sum D where C > 1000" → `=SUMIF(C:C,">1000",D:D)`

50–84%

Acceptable

Formula is likely correct but review before using. Prompt had some ambiguity.

- Prompt was partially specific
- Condition implied but not stated
- Formula type inferred from context
- Minor Word Radar warnings

Score 71%: "calculate bonus for sales" → `=A1*1.15` (assumed 15%)

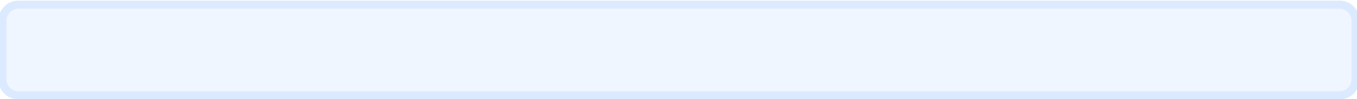
0–49%

Low Confidence

Formula may be incorrect. Revise prompt with specific column names and values.

- Prompt was too vague or general
- No columns, values, or conditions given
- Multiple interpretations possible
- Word Radar flagged unclear terms

Score 34%: "do something with data" → best-guess formula only





Installation Guide

4 steps to load SniperSheet in Microsoft Excel

1

Download manifest.xml
Open your browser and go to the manifest URL:

2

Open Excel Add-ins Manager
In Microsoft Excel: click Insert → Add-ins → My Add-ins

3

Upload manifest.xml
Click "Upload My Add-in" → select the manifest.xml file.

4

Open Smart Hub
Click the SniperSheet tab in the Ribbon → click



System Requirements

Minimum requirements to run SniperSheet

Requirement	Value	Notes
		Office 365 recommended for full features
		Required for Office.js task pane add-ins
		Offline local engine works without internet
		Handles the task pane embedded UI
		Task pane needs at least 400px sidebar

Manifest Download URL:

<https://8e832e48-8f9e-4168-9828-29c19ce7accc-00-12f81e1kjeof1.picard.replit.dev/api/addin/manifest.xml>
Generated dynamically — always returns the latest version



Frequently Asked Questions
8 common questions about SniperSheet answered

Yes. Groq free tier (Llama 3.3 70B, 30 RPM). The API key is on the developer's server. Users need no Groq account and pay

No. All AI requests route through a US Replit server. Users connect to Replit's HTTPS endpoint, never directly to Groq. No

3-model cascade: if primary (Llama 3.3 70B) fails, it retries with Llama 3 70B, then Llama 3 8B. If all fail, the local engine

Yes — partially. The local engine handles 35+ patterns (SUM, IF, VLOOKUP...) offline at zero latency. AI features need internet.

Excel 2016 and later on Windows and Mac. Office 365 recommended. Uses Office.js Manifest v1.1 with

Yes. Arabic input is fully supported in Smart Hub. Type in Arabic and SniperSheet returns the formula with an Arabic explanation.

Sent over HTTPS to Replit, which forwards only the text prompt to Groq. No data stored. History exists only in browser session

Yes. ReadWriteDocument permission lets SniperSheet write formulas directly into the active cell. Click "Apply to Cell" after



Developer Profile
The creator of SniperSheet

Mustafa Alsahlany

Developer & Designer — SniperSheet Excel Add-in

React, Express.js, TypeScript, Groq AI — Built on Replit

Iraq · 2026

Replit — Active

Groq AI — Free Tier

v1.0.0

April 2026

Technology Stack

Frontend

React 18 + Vite + TypeScript

Backend

Express.js + Node.js 24

AI

Groq SDK — Llama 3.3 70B

Styling

Tailwind CSS + shadcn/ui

Build

esbuild — CJS bundle

Packages

pnpm Monorepo Workspace

Technology Stack

Manifest

Office.js v1.1 Task Pane

Permission

ReadWriteDocument

Locale

ar-SA (primary) + en-US

Hosting

Replit Autoscale (US region)

Security

HTTPS / TLS 1.3

Icons

16 · 32 · 64 · 80 px PNG

Groq AI

Llama 3.3 70B · Free Tier · 30 RPM

3-Model Cascade · LPU Hardware

Replit

Autoscale · US Region · HTTPS

Node.js 24 · Express.js · API Proxy

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